

Auditing a Released Back-Translation Evaluator for Sign Language Production: A Public-Recipe Sufficiency and Replay-Probe Reproducibility Audit

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Abstract

Back-translation (BT) evaluators for sign language production (SLP) are learned measurement models: their rankings may reflect both the evaluated outputs and the inductive biases of a particular checkpoint. We ask whether a stress-test ordering observed under a released benchmark checkpoint reproduces from its publicly documented training recipe. On the 641-sequence PHX-public subset of PHOENIX-2014T, a constructed whole-donor replay probe scores 23.02 sacre-BLEU under the released SLRTP2025 evaluator versus 12.78 for the recorded test poses (gap **+10.24**, 95% CI [**+8.88**, **+11.62**]). Three findings follow.

First, the public recipe does not reproduce the released evaluator’s competence. Fourteen recipe-conditioned reconstructions reach development-set free-decode BLEU 6.5–9.4 (vs. released 13.38), training-set teacher-forced token accuracy 55–60% (vs. 96.5%), and training-set free-decode BLEU 7.2–9.3 (vs. 78.8). The released evaluator’s training-pool free-decode readout of 78.8 BLEU (70.7% exact match) far exceeds any constructible checkpoint (family max 9.3). A small-sample overfit test (BLEU reaches 100.0 on 50–100 samples) rules out a fundamental code-path defect.

Second, no constructible checkpoint reproduces the stress-test ordering. Fourteen same-architecture reconstructions yield all-negative probe-reference gaps (seed-level 95% prediction interval [**−1.94**, **−0.46**]); 38 non-degenerate heterogeneous-family checkpoints extend the negative range to [**−2.01**, **−0.21**]. Cross-evaluator re-decoding of a realistic noisy-gloss retrieval output yields 38/38 negative gaps. Third, on the matched 461-item subset of published human sign-to-text back-translations, reference replacement attenuates the gap by 90.5% (bootstrap CI [**7.44**, **10.86**] absolute; [**0.81**, **1.01**] ratio). Donor-pool substitution reverses the gap sign. The source-conditioned retriever shares surface-level vocabulary with

the original reference, so the attenuation reflects both reference-text properties and disruption of this lexical coupling.

The released evaluator’s readout-with-generalization signature—high training-pool fit (78.8 BLEU) plus retained in-domain performance (13.38/12.78)—is not reproduced by any constructed checkpoint. Permutation and held-out controls exclude pose-ignorant lookup as the explanation. These findings evidence public-recipe insufficiency for reproducing the released evaluator’s competence and its behavior on a constructed replay probe; they do not establish a causal checkpoint effect or general leaderboard instability.

Keywords: sign language production, back-translation evaluation, reproducibility audit, reference-frame sensitivity, learned evaluation metrics, PHOENIX-2014T

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-recipe sufficiency audit*: whether the released SLRTP2025 BT evaluator and its replay-probe reversal can be reproduced from the publicly documented training recipe. We address three research questions:

- RQ1. Can the public recipe reproduce the released evaluator’s competence and training-pool readout?
- RQ2. Does a frozen replay stress probe’s ordering reproduce across evaluator checkpoints constructed from the public recipe?
- RQ3. How sensitive is the stress-probe reversal to the reference text and to donor origin?

We do *not* frame the work as a test of checkpoint identity, nor as a direct test of the Saunders et al. (2021) rank-stability proposition, which is conditioned on similar model configurations: every checkpoint we can construct from the public recipe is less competent than the released evaluator (development-set free-decode BLEU-4 6.5–9.8 under the uniform beam-3 protocol vs. 13.38; training-log max 10.02), so competence and checkpoint identity are confounded and the non-reproduction result is scoped to the public recipe.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai (2025); Jiang et al. (2025), yet BT remains central to SLRTP2025 Walsh et al. (2025). Chen et al. (2022) showed that learned BERT-based evaluation metrics can fail to reproduce due to undocumented preprocessing, missing code, and baseline implementation differences—a methodology directly relevant to our finding that public artifacts are insufficient to reproduce a released checkpoint. An

unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by a retrieval–reference reversal on PHX-public: under the released evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale) versus 12.78 for the recorded test poses—which neither establishes better signing nor invalidates BT evaluation in general, but asks whether the ordering depends on properties of the released evaluator that the public recipe does not recover.

We address the three research questions with the following evidence. For RQ1: fourteen reconstructions reach dev free-decode BLEU 6.5–9.4 (vs. released 13.38) and training-set teacher-forced token accuracy 55–60% (vs. 96.5%); a small-sample overfit test (BLEU 100.0 on 50–100 samples) excludes code-path defects. For RQ2: the 14 reconstructions give a seed-level prediction interval of $[-1.94, -0.46]$; 24 heterogeneous-family checkpoints extend the range to $[-2.01, -0.21]$. For RQ3: reference replacement attenuates the gap by 90.5% on the matched 461-item subset (bootstrap CI $[7.44, 10.86]$ absolute; $[0.81, 1.01]$ ratio); donor-pool substitution reverses its sign. The probe’s source-conditioned retriever shares surface vocabulary with the original reference (§3.2), so attenuation reflects both reference-text properties and disruption of this lexical coupling. The released evaluator’s training-pool readout of 78.8 BLEU (70.7% EM) far exceeds any constructible checkpoint (family max 9.3).

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator’s training pool, so the probe isolates a risk that production systems face in diluted form; methodologically, the audit contributes a transferable template—provenance table, checkpoint registry with SHA-256 deduplication, evaluator-replication controls, and teacher-forced-vs-free-decode diagnostics—for public-recipe reproducibility testing; the specific reversal is a PHX-public released-evaluator case study and does not support general claims about BT evaluation or retrieval-based SLP.

2 Related Work

Learned evaluation in sign language production.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); BT was proposed as an SLP evaluation protocol by Saunders et al. (2021), who argued it shifts absolute scores but not relative rankings between similar configurations — a proposition our audit does not directly test (its similarity condition is unmet in our family) but re-examines for a released evaluator on which a retrieval-probe reversal appears. SignBLEU Kim et al. (2024), SiLVERScore Imai (2025), and pose-native measures offer complementary views, but BT BLEU remains central to SLRTP2025 Walsh et al. (2025).

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025’s leading entry is retrieval-based, making donor copying pertinent Walsh et al. (2025).

Memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020); a shortcut under one learned evaluator does not show the ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) show BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low recorded-pose BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU; and model-based generation metrics are further criticized by Yazdani et al. (2026); Cory et al. (2025); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2025) released human back-translations and documented reference information loss, Alkain et al. Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora, and Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Reproducibility of learned evaluation metrics.

Chen et al. Chen and Eger (2022) showed that reproducing BERT-based evaluation metrics can fail due to undocumented preprocessing, missing code, and baseline implementation differences, even when model weights are released—a finding methodologically close to our public-recipe non-reproduction result. The difference is that our audit targets a full BT evaluator pipeline (architecture, training recipe, and checkpoint) rather than a metric alone, and the non-reproduction is demonstrated across 38 non-degenerate independently trained instances.

PHOENIX-2014T as a language resource: known limitations.

PHOENIX-2014T Camgoz et al. (2018) is the most widely used parallel corpus for German Sign Language (DGS) production and translation, containing 8,257 weather-forecast interpretations by nine signers aligned with German speech transcriptions. Its broad adoption makes auditing its evaluation infrastructure relevant to the language-resource community, but several properties constrain the audit’s generalizability. First, the corpus is template-dense: weather forecasts reuse a restricted vocabulary and stereotyped syntactic patterns, inflating n-gram overlap between training and test sets Alkain et al. (2026). Our stratified analysis confirms this: the 32 items whose reference exactly matches a training text account for a disproportionate share of the corpus-level gap. Second, the reference texts are speech transcriptions, not translations of the signing—they capture spoken German, not what the signer expressed in DGS. Czehmann et al.’s human back-translations quantified the resulting information loss Czehmann et al. (2025). Third, the corpus uses a single-channel upper-body pose representation (~ 130 joints), which discards handshape detail critical for sign recognition. Fourth, the signer pool (9 signers) is small, and signer-dependent evaluation can overestimate translation capability Artiaga et al. (2025). These limitations are not unique to PHOENIX-2014T—many sign language corpora share template density, speech-transcribed references, and limited signer diversity—but the specific

lexical coupling between source and reference (the German sentence is both input and evaluation target) amplifies their effect in our stress probe. Section 3 documents our provenance controls; the CSL-Daily boundary check (SI Sup. L) tests whether the readout pattern generalizes to a non-German corpus with different structural properties. The transferable components of our audit template—provenance table, checkpoint registry, leakage controls—are designed to apply to any learned evaluator paired with a publicly released training corpus.

Table 1: Novelty relative to closely related PHOENIX-2014T and learned-metric reproducibility studies.

Study	Primary finding	Method	Distinction from present work
Czehmann et al. (2026)	Reference replace-ment changes BLEU scores and rankings on PHOENIX-2014T	Human back-translations of test set	We apply the matched-subset protocol to the released evaluator with a frozen replay probe and quantify paired attenuation
Alkain et al. (2026)	Highest train–test text overlap among tested SLT corpora	N-gram overlap quantification	We use their finding to contextualize why PHX-public is particularly susceptible to reference coupling
Artiaga et al. (2025)	Signer-dependent evaluation overestimates SLT capability	Signer-disjoint splits	We control for signer identity in per-item decomposition (signer contributes $\leq 1\%$ of gap variance)
Jiang et al. (2025)	BT likelihood is a more stable correlate of decoded BLEU than human judgment	Meta-evaluation of pose/embedding/BT metrics	We report that teacher-forced likelihood does not reproduce the reversal (REC NLL 3.060 vs PURE 3.479)
Chen et al. (2022)	BERT-based metrics fail to reproduce due to missing preprocessing/code	Multi-implementation replication	We extend the learned-metric reproducibility question to a full SLP evaluator pipeline with 70 training runs
This work	Public recipe does not reproduce the released evaluator’s competence or replay-probe behavior	70 training runs across 13 families, fixed-output cross-evaluator re-decoding, reference/donor substitution	—

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint, x_n the source sentence (German) for item n , and $o_{i,n}$ the pose sequence for output condition i and item n ; E_c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^N)$, where y_n is the reference text and M is the

Table 2: Three-way data provenance for PHX-public. The released evaluator’s training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs share the reason `absent_from_slrtp2025_released_pose_records` (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (.pt)	7,060	515	641
Released evaluator <code>config.yaml</code> manifest	7,060	515	641
Missing from official	36	4	1

corpus-level sacreBLEU functional unless stated otherwise. We call a result *evaluator-pipeline-sensitive* when a substantive pairwise ordering changes across independently trained checkpoints under a fixed output set. The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval-generator compositions; oracle-gloss variants are diagnostic controls.

Operational definition of public-recipe sufficiency.

A publicly documented training recipe is *sufficient* for reproducing a released evaluator when checkpoints constructed from it reach a predefined competence gate: $|\Delta_{\text{dev BLEU-4}}| \leq \delta_{\text{BLEU}}$ and $|\Delta_{\text{dev WER}}| \leq \delta_{\text{WER}}$, with $\delta_{\text{BLEU}}=1.0$, $\delta_{\text{WER}}=3.0$ (released dev values: 13.38 / 83.37). These thresholds are post-hoc—no prior standard exists for BT-evaluator reproducibility—but the gate conclusion is insensitive across a $\pm[0.5, 2.0]_{\text{BLEU}} \times \pm[1.5, 4.5]_{\text{WER}}$ grid (§4.2). Insufficiency can arise from two sources. (i) *Information incompleteness*: the public artifacts omit training details (data preprocessing specifics, learning-rate schedule shape, validation-selection protocol, intermediate checkpoints) that the original training relied on. (ii) *Stochastic underspecification*: a single rare seed or optimizer trajectory produced the released result. Our evidence favors (i): the competence shortfall is consistent across 14 seeds (dev BLEU-4 6.5–9.4 vs. 13.38; training-set teacher-forced token accuracy 55–60% vs. 96.5%), and local-robustness checks confirm the released checkpoint is not a fragile singular point (SI Sup. F). Without the original training command and intermediate checkpoints, the two sources cannot be fully separated.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 2 gives the three-way provenance: the official PHOENIX-2014T split, the released pose materialization, and the released evaluator’s actual training manifest (confirmed via `config.yaml`, whose manifest matches the shipped .pt tensors). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L); the protocol summary is in SI Table S28.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution; a three-path regression confirms no double subsampling and that the official starter-bundle Progressive Transformer reproduces the official 1.5942 BLEU under our pipeline (details in SI Sup. F). PHX-public uses $\alpha=0.88$ (selected on dev); the supplementary splits use $\alpha=0.5$.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is 9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbafa301. Reproduction is supported at three layers: (i) a single command (`make core-audit`) regenerates the registry and every headline number from the committed per-item decodes; (ii) each analysis script re-executes from the committed per-checkpoint data and hypotheses; (iii) full retraining and re-evaluation from the public recipe is scripted (`scripts/train_reconstruction.py`, `launch_reconstructions.sh`), requiring the training bundle. An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N: 610/641 items (95.2%) are identical across materializations, and a 14-reconstruction comparison gives mean $|\Delta\text{gap}| = 0.65$ BLEU (max 1.45), so qualitative conclusions are unchanged.

Lexical coupling of the source-conditioned probe.

In PHX-public, the source sentence x_n (German weather report) is identical to the reference text y_n^{orig} at the sequence level (the reference is a speech transcription of the same German sentence). The TN-PURE-v1 retriever selects training-pool donors by token-set Jaccard similarity between x_n and training texts, so it preferentially retrieves donors whose accompanying training texts share surface-level vocabulary with both the source *and* the reference. Scoring a donor copy against this lexically coupled reference under a learned evaluator therefore confounds donor similarity, evaluator readout, and reference overlap. This coupling is deliberate in the stress-probe design—it isolates the strongest form of the shortcut—but it means the human-reference attenuation analysis (§4.2.1) reflects disruption of the lexical coupling as well as reference-text information loss, not a clean separation of the two.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with `::2` subsampling to 12.5 fps, preserving the donor’s original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α -blending of upper-body joints ($\alpha=0.88$; full

joint-set definitions in the artifact). The matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the retrieval advantage co-occurs with donor origin, we construct frozen probe systems from the same 641 queries: **UNSEEN-PURE-v1** retrieves from the 640 other test poses; **CROSSFIT-PURE-A/B-v1** split queries by hash into two folds, each retrieving from the other fold; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ± 0.1 Jaccard of each UNSEEN donor (622/641 admit one); **SEEN-RAND640-MATCHED-v1** applies matched selection inside frozen 640-donor sub-pools. SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry; these are constructed probes, not deployable generators (construction rules in the artifact).

3.3 Evaluator checkpoint family

The primary inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and six *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture on a common frozen partition (primary seeds 101–606; eight extension seeds 707–1405 use the identical protocol). The original bundle’s `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈ 102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.2.1, and intermediate checkpoints are not published. We use two reconstruction protocols — *paper-derived* (14 runs, legacy script, 300 epochs, NLL selection) and *released-config* (4 runs, up to 3,000 epochs, decoded-BLEU early stopping) — sharing the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), `[:: 2]` subsampling, Adam (1e-3, wd .001), batch size 256, and plateau $\times 0.8$ LR schedule (full provenance in SI). In total 70 runs across 13 families (registry in SI Sup. D); 43 were decoded on PHX-public under the canonical §3.2 donor registry, corresponding to 42 unique weight binaries — the long-schedule seed-202 artifact is byte-identical to the config-faithful seed-202 artifact (identical `config.yaml` and wandb run, identical per-step losses, identical best-checkpoint SHA-256), so it is a duplicate of that run rather than an independent replicate. Of the 43 decoded runs, five are degenerate (three $\alpha=1.0$ distillation students with empty hypotheses, two smallest ladder fractions with BLEU ≈ 0) and are excluded, so the non-replication count is 38 non-degenerate decoded runs (37 unique binaries), all with strictly negative gaps.

Training-pool readout (descriptive observation).

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match, while retaining dev/test performance of 13.38/12.78. The constructible family’s readout is far lower (reconstructions: 7.2–9.3 BLEU, 1.5–2.3% EM; distillation: 6.6–8.9 BLEU, 1.3–1.9% EM); no constructible checkpoint approaches both simultaneously. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$ vs. reconstruction family: 0.93–1.12) quantifies how much the model reads out above its in-domain performance. We report these as continuous descriptive observations: the released evaluator was observed first, the family range afterwards, and any threshold is post-hoc.

3.4 Reconstruction sanity checks

Because all 70 training runs share the same code, data preprocessing, and hyperparameters, we performed three checks to rule out a systematic implementation defect shared across all runs:

- Small-sample overfit test.** We trained a reconstruction model on 50 (and separately 100) randomly sampled training sequences for ≈ 200 epochs with the identical pipeline. On the 50-sample set, teacher-forced NLL fell from 7.66 to 0.009 and free-decode sacreBLEU reached 100.00; on the 100-sample set, free-decode BLEU also reached 100.00. The reconstruction code can therefore overfit small data to near-perfect free-decode BLEU, ruling out a fundamental defect in checkpoint loading, tokenizer, padding-mask, EOS, loss normalization, or the beam-search decoding path (results: `results/overfit_test.json`).
- Inference regression.** Released and reconstruction checkpoints are loaded through the identical `make_back_translation_model()` function and decoded with the identical `back_translate()` pipeline. All 15 checkpoints (released + 14 reconstructions) share byte-identical `txt.vocab` and `gls.vocab` files (confirmed by SHA-256). The released checkpoint decoded through this path reproduces the canonical REC value exactly (12.78 on 641 items; `results/inference_regression.json`).
- Free-decode diagnostics.** Table 3 reports per-checkpoint teacher-forced token accuracy, NLL per token, and free-decode BLEU on training and development sets. The released evaluator achieves 96.5% training-set teacher-forced token accuracy and 78.8 free-decode training BLEU (full-pool, beam-3); the fourteen reconstructions achieve 55.0–59.6% (mean 57.2%) token accuracy and 7.2–9.3 (mean 8.35) free-decode training BLEU. The reconstructions achieve slightly lower development-set teacher-forced NLL (2.63–2.68 vs. released 3.24), consistent with less overfitting, but their teacher-forced token accuracy is comparable on dev (45.8–47.4% vs. released 48.6%). The gap between released and reconstructed free-decode BLEU is therefore present in both training and development data and is already visible in teacher-forced token-level metrics, not only in free-decoding.

Table 3: Teacher-forced and free-decode diagnostic statistics for the released evaluator and the fourteen reconstruction seeds (full table in `results/checkpoint_stats.json`). Teacher-forced values are on 500 randomly sampled items; free-decode values are full-pool (7,060 train / 515 dev) under the uniform beam-3 protocol (§3.4). sacreBLEU signature: BLEU|c|n:4|e:no|tok:13a|s:exp|v:2.5.1.

Checkpoint	Train tok. acc.	Train NLL/tok	Train free-decode BLEU	Dev NLL/tok	Dev free-decode BLEU
Released	0.965	0.158	78.82	3.236	13.38
Reconstructions (range)	0.550–0.596	1.72–1.92	7.20–9.25	2.63–2.68	6.49–9.37
Reconstructions (mean)	0.572	1.82	8.35	2.66	8.13

3.5 Discovery timeline and pre-registration status

The +10.24 reversal was first observed on 2026-07-24 in the canonical materialization with exact-normalized-text exclusion (SHA-256 9170a530); an earlier pre-exclusion materialization (gap +11.01) had been noted on 2026-07-22. The six primary reconstruction seeds (101–606) were specified before their training results were inspected, and the eight extension seeds (707–1405) were added post-hoc after observing the six primary results. The Jaccard-based retriever, PURE donor-copy protocol, exact-normalized-text exclusion rule, and 641-query set were locked before reconstruction training began, after exploring approximately four retrieval strategies (Jaccard, LaBSE, multi-SBERT, oracle gloss) and three construction variants (pure copy, PT-composed, oracle-gloss-composed) on the released evaluator alone. No external pre-registration exists for the specific +10.24 probe; the six primary seeds constitute the only pre-specified confirmatory evidence, and all heterogeneous-family results are post-hoc descriptive exploration. The full registry of which family enters which claim is in SI Sup. C.

3.6 Metrics and uncertainty

The recorded test poses (REC) are decoded by each BT recognizer; we avoid the abbreviation “ground truth” because the recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale as the primary metric Post (2018) (full signature: BLEU|c|n:4|e:no|tok:13a|s:explv:2.5.1); sentence-level statistics (pass-through, slot-F1) are reported separately on the 0–1 scale. BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020) (**bert-base-multilingual-cased**, no baseline rescaling, F1), and WER (jiwer 3.1.0 normalized) provide complementary decoded-text *cross-metric consistency* checks within the same BT hypotheses—their consistent direction across evaluators is not independent semantic validation because all metrics share the identical decoded outputs Yazdani et al. (2026). BT likelihood and pose-DTWp provide non-decoded controls; a non-standard BLEURT substitute used only for a secondary reference-sensitivity check is documented, with its validation against canonical BLEURT-20 and its known bias, in SI Sup. O.

Uncertainty is quantified with paired sequence-level bootstrap (10,000 resamples, seed 42, percentile 95% CIs) over per-item n-gram sufficient statistics, re-aggregated into a corpus BLEU per resample. Because queries reuse pose donors (565 unique donors for 641 queries), donor-cluster and query×donor two-way cluster bootstraps complement the item bootstrap (SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. A descriptive seed-level 95% Student-*t* prediction interval is reported for the recipe-conditioned reconstruction population; its interpretation is strictly “given a future random seed under this identical legacy implementation,” not a claim about all public-recipe implementations. The eight extension seeds are post-hoc additions and enter only this interval (never the six primary comparisons); splitting by pre-registration status, the six primary seeds give PI [−2.48, −0.28] and the eight extension seeds [−1.61, −0.54], all fourteen leave-one-seed-out folds keep the interval below zero, and a two-level seed×item bootstrap (seeds resampled with

replacement, then items within each drawn seed) gives CI $[-1.49, -0.93]$ (SI Sup. B). The single primary estimand is the PURE-REC decoded-sacreBLEU gap under the released evaluator; all other contrasts are secondary and are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

3.7 Terminology: run, checkpoint, unique binary, family, evaluator

Throughout, *run* denotes a single training execution; *checkpoint* denotes the selected model weights from that run; *unique weight binary* counts distinct SHA-256 hashes of checkpoint files (the cf.202 and ls.202 runs produced byte-identical weights from the same wandb run); *family* denotes a group of runs sharing a training protocol; and *evaluator* denotes a checkpoint used to decode fixed outputs. The 14 paper-derived reconstruction runs use NLL selection on the legacy script; the 4 released-config runs use decoded-BLEU early stopping. Both share the documented architecture and hyperparameters (§3).

4 Results

4.1 Stress probe and conditioning controls (RQ2 preamble)

Under the released SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses—a retrieval-recorded-poses gap of +10.24 BLEU points (95% CI $[+8.88, +11.62]$; 10,000 sequence-level resamples). This fixed-checkpoint reversal motivates the analysis but does not indicate superior sign production.

Controls localized the observation to source-conditioned donor retrieval (Table 4): pure donor copy exceeded the oracle-gloss composed variant (11.8), canonical PT-composed (18.86), PT baseline (1.59), and 20-seed random-donor baselines (0.90 PURE, 0.77 COMP), while oracle-gloss retrieval (leakage) did not exceed REC (11.4–11.8 vs 12.78). Non-decoded controls do not reproduce the reversal: teacher-forced NLL gives REC 3.060 vs PURE 3.479, COMP 3.411, PT 4.074, and normalized pose-DTW shows REC 0 by construction vs PT/PURE/random 0.00678–0.00830. Complementary text metrics agree on the same hypotheses (SI Table S22): sacreBLEU-4 +10.24, BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER −13.90, BERTScore-F1 +0.066; we therefore describe the observation as a *decoded-text-metric reversal* with sacreBLEU-4 as the primary decoded metric.

Decoding-parameter sensitivity.

Re-decoding across beam size $\{1, 3, 5\}$, length penalty $\{-1, 0\}$, and max length $\{30, 50\}$ leaves the PURE-REC gap at +9.9/+10.2/+10.8, so the reversal is not a decoding artifact; length penalty and max length are negligible.

Pure donor copy outperformed composition for every source-conditioned retriever, and source-text Jaccard was highest (23.02, vs LaBSE 19.2, multi-SBERT 16.5): the reversal needs no oracle-gloss leakage, and pure copy — not composition — is its strongest form.

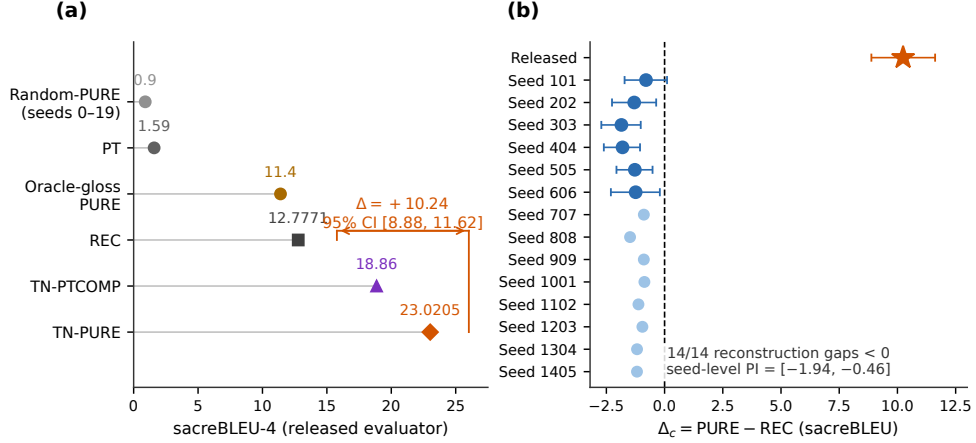


Fig. 1: Headline reversal under the released evaluator and its absence across recipe reconstructions. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, 95% CI [8.88, 11.62]), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE–REC gaps: released evaluator (+10.24) vs. fourteen recipe reconstructions, all negative (paired-item bootstrap 95% CIs; seed-level 95% PI [−1.94, −0.46]). All reconstructions are less competent than the released evaluator; the figure demonstrates public-pipeline non-replication, not an isolated causal checkpoint effect.

Table 4: PHX-public conditioning and construction controls (641 sequences). Columns distinguish whole-donor copy from PT-scaffold composition; all rows share the query set, donor pool, evaluator, and sacreBLEU protocol. Semantic retrievers (LaBSE, multi-SBERT) lie between the canonical retriever and oracle gloss (artifact).

Conditioning	Pure donor copy	PT+retrieval composed
Source text (German Jaccard)	23.02	18.86
Oracle gloss (input leakage)	11.4	11.8
Random donor (seeds 0–19)	0.90 (SD 0.24)	0.77 (SD 0.19)

4.2 RQ2: Multi-checkpoint reproduction of the stress-test ordering

TN-PURE-v1 yields an original-evaluator gap of +10.24 BLEU points (95% CI [+8.88, +11.62]) and TN-PTCOMP-v1 +6.09 [+4.76, +7.41]; across all fourteen reconstructions both gaps are negative (PURE range [−1.85, −0.80]; seed-level 95% prediction interval [−1.94, −0.46], mean −1.20, SD 0.33, 0/14 positive; the six pre-registered primary seeds alone give [−2.48, −0.28] and the eight extension seeds

$[-1.61, -0.54]$, with the interval below zero in all fourteen leave-one-seed-out folds), and for the six primary runs original-minus-run difference-in-differences estimates range from +10.89 to +11.99, every interval excluding zero.

4.2.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 5), and all evaluators are decoded with the canonical §3.2 donor registry (exact-normalized-text exclusion applied).

Table 5: Canonical fixed-output evaluation on the released 641-pose PHX-public subset, decoded with the §3.2 donor registry. PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1. Gaps are relative to local REC (BLEU points). Per-cell values in the artifact.

Evaluator	REC	PURE	Gap	COMP	Gap
Original	12.78	23.02	+10.24	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16
Extension 707–1405 (range)	8.05–9.87	6.86–8.97	−1.48 to −0.87	6.90–8.83	−1.03 to −0.44

Competence is assessed on the released 515-sequence development subset, never on PHX-public. Original obtains dev BLEU-4 13.38; under the uniform full-pool beam-3 free-decode protocol the 14 reconstructions obtain dev BLEU-4 6.5–9.4, and their selected validation NLL (2.641–2.712) is lower than Original’s teacher-forced dev NLL (3.235), so likelihood and decoded recognition competence are not interchangeable. Consequently, competence and unavailable original-training details remain confounded with evaluator identity; the experiment identifies evaluator-pipeline sensitivity, not a causal checkpoint-identity effect.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both $|\Delta \text{dev BLEU-4}| \leq 1.0$ and $|\Delta \text{dev WER}| \leq 3.0$ (released dev values 13.38 / 83.37). None of the 52 non-distillation trained checkpoints passes the gate. Under the uniform full-pool beam-3 protocol, the 26 non-degenerate runs among the 28 with released weights leave dev BLEU-4 short by 3.6–6.9 and dev WER exceeds the 86.37 bound by 0.5–10.6 (the two smallest ladder fractions decode empty); the 24 runs whose weights the artifact withholds log dev BLEU-4 at most 10.02 under the original selection protocol, failing the BLEU half by at least 3.4 at every grid width. The nine released-weight fine-tunes (SI Sup. F) all pass the gate (dev BLEU-4 12.92–13.18, dev WER 84.02–84.40), but they perturb

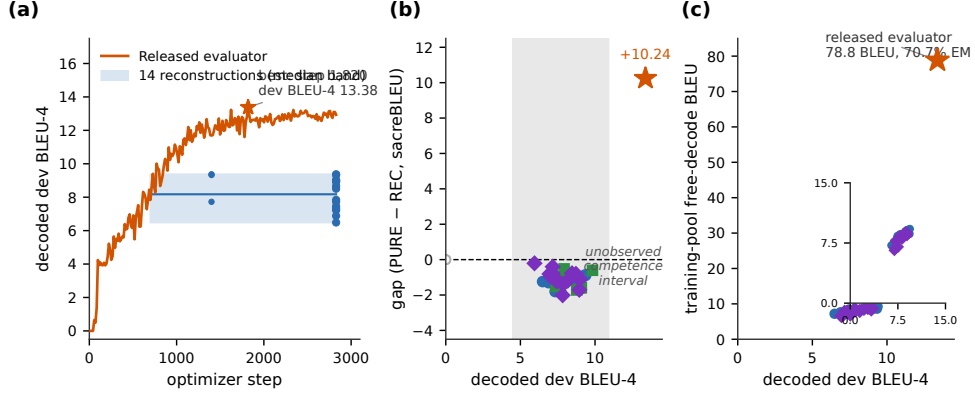


Fig. 2: Decoded competence and training-pool readout separate the released evaluator from every constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (archived validation log, 202 logged points); fourteen reconstructions form a median band (blue), plus uniformly decoded epoch-25/50 points. (b) Canonical PURE-REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), config-faithful/step-faithful/confirmation/long-schedule (squares), rescue (triangle), distillation students (diamonds), released evaluator (star); the gray band marks the observed 6.5–9.8 range (uniform protocol) and the (9.8, 13.38) interval is unobserved by recipe-constructed checkpoints (the nine released-weight fine-tunes occupy 12.92–13.18 but are weight perturbations); degenerate checkpoints are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses) vs. decoded dev BLEU-4: every constructible checkpoint reads out 6.6–9.3 BLEU, while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38. Families are descriptive; checkpoints are not IID replicates.

the released weights rather than constructing checkpoints from the recipe, so they are excluded from the sufficiency assessment. The rescue search (batch size, dropout, weight decay, label smoothing, longer budget; SI Table S29) also fails: its best variant (weight-decay 0, seed 202) yields REC 9.88 / PURE 8.29 (gap -1.59). The gate conclusion is threshold-insensitive: across a $\pm[0.5, 2.0]$ BLEU $\times \pm[1.5, 4.5]$ WER grid, 0 of 28 uniform-decoded runs pass.

What the released training log establishes.

The bundle’s `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 2a), while all constructible reconstructions plateau at 8–10 by epoch 50–100; the training dynamic behind this slope is not explained by public artifacts. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at $k=8$: REC 8.75, PURE 11.69, gap $+2.94$; SI Sup. F).

Cross-metric consistency across checkpoints.

Table 6 reports the PURE–REC gap on the same fixed output under each of the seven primary evaluators across six metrics (WER and NLL sign-flipped so positive means PURE better). Under the original evaluator, the gap is positive and large for all decoded-text metrics, while teacher-forced NLL is more negative for PURE than REC (−0.419 sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under the six reconstructions the reversal disappears: all chrF and ROUGE-L gaps are negative, all WER gaps favor REC, and sacreBLEU-4 gaps are all negative (range [−1.85, −0.80]). The reversal is therefore a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4.

Table 6: PURE–REC gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding). chrF, WER, and ROUGE-L use the official SLRTP2025 implementations. WER and NLL are sign-flipped so positive means PURE is better than REC (lower WER, lower NLL). BERTScore-F1 uses `bert-base-multilingual-cased` (no baseline rescaling). The gap is consistently positive under the original evaluator and consistently near-zero or negative under the six primary reconstructions; the eight extension seeds show the same pattern (see main-text multiseed factorial table).

Evaluator	Δ BLEU-4	Δ BLEU-1	Δ chrF	Δ WER [†]	Δ NLL [†]	Δ BERTSc.
Original	+10.24	+20.94	+12.93	+13.90	−0.419	+0.066
Seed 101	−0.80	−1.23	−1.44	−2.95	−0.219	−0.005
Seed 202	−1.30	−1.33	−1.57	−3.52	−0.233	−0.006
Seed 303	−1.85	−1.99	−2.26	−1.53	−0.240	+0.001
Seed 404	−1.81	−1.60	−1.90	−1.87	−0.194	−0.001
Seed 505	−1.28	−1.53	−1.59	−3.41	−0.268	−0.004
Seed 606	−1.24	−1.26	−1.55	−3.46	−0.169	−0.001

[†]WER and NLL are sign-flipped: positive means PURE is better than REC.

Rank stability (see SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean τ_b =0.44, ρ =0.57).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2025) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across

sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a paired attenuation of 9.08 BLEU points (90.5%, bootstrap CI [7.44, 10.86] on the absolute attenuation and [0.81, 1.01] on the attenuation ratio; SI Table S31; SI Fig. S5; per-item paired view in SI Fig. S2), and the residual gap is not distinguishable from zero (paired item CI [-0.13, 1.99]). Teacher-forced BT likelihood does not reproduce the reversal under either reference (REC NLL 3.060 vs PURE 3.479 on original refs; 3.41 vs 3.71 on human refs) Jiang et al. (2025). Scoring against both references jointly (closest-length convention) leaves the gap at +11.50 — *larger* than original: TN-PURE donors are selected by lexical overlap with the source sentence, and the original reference is a speech transcription of that same sentence, so adding it injects a second lexically coupled target the replay decode can match; the gap inherits the lexical coupling rather than averaging it away. The reversal is thus specific to reference frames containing the original speech-transcribed register. The full-641 comparison (+10.24 $\rightarrow$ +0.86, 91.6%) mixes reference replacement with floor effects from low-confidence items. This is a *descriptive contrast of two corpus gaps*, not a causal attribution: the attenuation cannot distinguish reference-text defects, paraphrase/translationese shift, BLEU floor effects, disruption of the lexical coupling, or evaluator training-distribution alignment. Under all six reconstruction checkpoints the human-reference gap is negative (-0.15 to -0.65), and the asymmetry is consistent across metrics (chrF 82.2%: +12.73 $\rightarrow$ +2.27; BLEU-1 84.0%: +20.01 $\rightarrow$ +3.19; SI Table S23), including a BLEURT-substitute check agreeing with Czehmann et al. (SI Sup. O). It quantifies how decoded-text scores shift when the reference frame changes, not whether the replay donor motion conveys target-equivalent meaning: the human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so the semantic adequacy of TN-PURE-v1 is left unassessed.

A natural concern is symmetric floor compression. Switching from original to human references, REC drops 7.59 BLEU-4 points (50.8%) but PURE drops 16.68 (66.8%) — a larger fraction (ratio 1.31 $\times$), and the asymmetry is even larger under BLEU-1 and chrF (SI Table S23); simple symmetric compression is insufficient, though the asymmetry does not discriminate between reference-text defects and training-distribution alignment.

4.3 Donor-origin contrast

If the released-evaluator reversal requires exposure to training-pool donors, retrieval restricted to unseen poses should remove or invert it: under the original evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU

vs. REC 12.78 (gap -4.27), a donor-pool substitution estimand of $+14.51$ BLEU points (SI Table S30). This is a *donor-pool substitution effect*, not a pure exposure effect: SEEN and UNSEEN differ simultaneously in pool size (7,060 vs 640), median donor-reference Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure. Cross-fit controls reproduce the direction (CROSSFIT pooled -3.8), and all reconstruction pool-origin contrasts lie in $[-0.15, +1.22]$.

Path-dependent decomposition and balanced factorial.

Telescoping-path decomposition (SI Sup. E) gives origin estimates of $+5.8$ to $+8.4$ BLEU across admissible orderings, because size and selection interact. The better-balanced 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity) estimates the pool-origin main effect at $+2.08$ [$+1.56, +2.57$], similarity at $+6.96$ [$+5.93, +8.08$], and their interaction at $+3.70$ [$+2.63, +4.79$] (all Holm $p=0.0003$); across reconstructions the seen main effect shrinks to $+0.51$ and the interaction is null (-0.39). Because seen donors remain more similar than unseen within strata (SMD $+0.4$), this is a train-pool vs test-pool provenance contrast, not an identified training-exposure effect.

4.4 Competitive-system context

Because the audit’s stress-test set is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: computed under the same released evaluator, its “Ground Truth” row reproduces our canonical REC numbers exactly (BLEU-4 12.78, BLEU-1 34.40, WER 85.77 Walsh et al. (2025); SI Table S25). The retrieval-based winner USTC-MoE sits roughly even with REC on the lexical BT metrics (BLEU-4 -0.72 , BLEU-1 $+0.45$, chrF $+2.24$; WER 93.49 vs 85.77) while its DTW-MJE (0.0448) and duration ratio (1.438) indicate considerable pose-level deviation from recorded signing — a single-evaluator co-occurrence that does not establish that the released evaluator favors retrieval entries; the two fixed-output cross-evaluator re-decoding studies below resolve this question directly.

Cross-evaluator re-decoding of realistic retrieval outputs.

Per-team pose outputs are not archived, so we re-decoded two fixed outputs of realistic retrieval pipelines under all 44 evaluator checkpoints (the released evaluator plus the 43 reconstructed runs; per-evaluator cells in SI Table S32–S33). First, the *noisy-gloss retrieval* pipeline — the deployment-realistic analogue of TN-PURE-v1, which predicts each test sentence’s gloss with a text-to-gloss NMT and retrieves the nearest training donor by predicted-gloss Jaccard (SI Sup. P) — decodes at 12.65 under the released evaluator versus 12.78 for REC (gap -0.13), reproducing the SI value: this realistic retrieval system is roughly even with recorded signing, unlike the constructed stress probe. Second, we rebuilt the official winner’s fixed test outputs from its released pipeline intermediates (text-to-gloss predictions, gloss-to-pose dictionary, official frame counts; 497 of 641 PHX-public keys; 471 retrieval and 26 copy branches; SI Sup. P): the rebuild decodes at 2.42 under the released evaluator (REC 12.36 on the same keys, gap -9.94), far below the officially reported leaderboard score of 12.06, so it is pipeline-faithful but not score-faithful to the archived submission. Under the 43 reconstructed

evaluators, every non-degenerate checkpoint gives a negative gap for both systems (noisy-gloss range $[-1.94, -0.07]$, mean -1.18 ; USTC-MoE range $[-8.30, -4.28]$, mean -7.17 ; the five degenerate checkpoints decode empty or near-empty hypotheses and give gaps of 0). Neither realistic system recovers a retrieval-favoring ordering under any reconstructed evaluator, and the released evaluator does not favor them either (gaps -0.13 and -9.94): the $+10.24$ reversal is specific to the constructed whole-donor probe. As a benchmark-organizer responsibility, archiving fixed per-team outputs alongside public scores would make rank-stability audits feasible for third parties.

Released-checkpoint local-robustness checks (experiments A1–A2, SI Sup. F).

Two perturbation designs test whether the reversal is a fragile singular point versus a locally robust property near the released checkpoint. Experiment A1 adds i.i.d. Gaussian weight noise at increasing scale σ : the reversal is preserved within ± 0.75 of $+10.24$ for $\sigma \leq 3 \times 10^{-3}$ (where dev BLEU-4 stays within 12.50–13.38) and collapses in lockstep with competence at $\sigma \geq 10^{-2}$. Experiment A2 fine-tunes the released checkpoint with three seeds at small learning rates (5e-6–3e-5, 60 epochs each), moving weights along real gradient directions rather than isotropic noise; all nine variants preserve the reversal (gaps $+10.37$ to $+10.77$, dev BLEU-4 12.92–13.18, training-pool readout 80.5–81.8 BLEU / 72.6–74.0% EM). Both designs confirm the reversal is locally robust near the released weights — not a fragile identity-level artifact — but they probe only the vicinity of the released checkpoint and do not demonstrate that an independently trained, competence-matched model would exhibit the reversal, nor do they narrow the unobserved competence interval. Full sweep and figures in SI Sup. F.

Per-item gap decomposition (experiment C, SI Sup. K).

An OLS regression ($R^2 = 0.33$, $n = 641$) attributes the largest share of per-item gap variance to high-overlap-train count, template density, and donor-reference Levenshtein distance; signer identity contributes $\leq 1\%$. Full model, coefficient table, and figure in SI Sup. K.

4.5 RQ1: Public-recipe insufficiency—what the released evaluator has that reconstructions lack

The non-reproduction in RQ2 raises the prior question of RQ1: can the public recipe reproduce the released evaluator’s competence? We approach it through five descriptive comparisons (Fig. 2b,c; Table 7; full version in SI Sup. D), reporting them as descriptive triangulation, not five independent tests.

Readout with generalization: the distinctive combination.

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match — strong readout — while retaining non-trivial dev/test performance (13.38/12.78). Under the identical full-pool, beam-3 free-decode protocol, the constructible family stays far below: 14 reconstruction seeds span train BLEU

Table 7: Summary of the five descriptive comparisons. The released evaluator sits far outside the family range on all five views. Per-view family composition in SI Sup. D.

Comparison	Family n	Released evaluator	Family range	Released – family max
(i) Gap–competence ratio	42	+10.24 at dev 13.38	$[-2.01, -0.21]$ (38 non-degenerate runs)	+10.45
(ii) Pass-through	24	3.09 (sentence)	1.05–1.31	+1.78
(iii) Train-pool read-out BLEU	27	78.8 (70.7% EM)	6.6–9.3 (1–2% EM)	+69.5
(iv) Frame-reversal loss	1	–16.1 (PURE)	–4.3 (wd0 rescue)	11.8
(v) Competence gate	52	13.38	6.5–9.8 (uniform; training-log max 10.02)	+3.55

7.2–9.3 (EM 1.5–2.3%) at dev BLEU 6.5–9.4, and 12 non-degenerate distillation students span train BLEU 6.6–8.9 (EM 1.3–1.9%) at dev BLEU 7.0–9.0 (training-time dev BLEU-4 under the original selection protocol reaches 10.9; SI Table S26, SI Sup. E). None of the 27 non-degenerate constructible checkpoints approaches 78.8/70.7%, and the released evaluator’s donor-content pass-through is a $2.4\times$ outlier (ratio 3.09 vs. family 1.05–1.31). Across the 31 checkpoints with uniform full-pool readouts (30 constructible, plus the released evaluator), training-pool readout tracks decoded competence (Spearman $\rho=0.916$, $n=31$, $p<10^{-12}$; $\rho=0.873$ over the 27 non-degenerate constructible checkpoints) and is uncorrelated with the train-minus-dev overfit indicator ($\rho=0.040$) or with the PURE–REC gap ($\rho=-0.082$; full table in SI Sup. D): within the constructible family the readout behaves as a competence proxy rather than an overfit signature, and the released evaluator’s readout-with-generalization combination is not explained by any within-family overfit–readout relationship.

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact: five controls (SI Sup. M) rule out four shortcut classes — teacher forcing (code-path audit confirms autoregressive beam search), output–reference misalignment (label permutation collapses BLEU from 78.6 to ~ 1.0), pose-ignorant ID/cache lookup (input-side pose permutation with verified output equivariance collapses BLEU to <1 ; a same-signer held-out control reaches only 12.19 BLEU / 3.2% EM on unseen poses), and narrow-text memorization (high exact-match across all 9 signers and unique texts). The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure.

Knowledge-distillation competence ladder.

To probe the unobserved competence interval above the reconstruction ceiling (dev BLEU-4 9.8 under the uniform protocol), we trained 15 distillation students from the released evaluator as teacher (five strengths $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, three seeds, $T=2$). All twelve $\alpha<1.0$ students have negative PURE–REC gaps (–2.01 to –0.77) with dev BLEU-4 up to 9.0 under the uniform protocol (training-time dev BLEU-4 up

to 10.9 under the original selection protocol) and readout 6.6–8.9 BLEU / 1.3–1.9% EM; the three $\alpha=1.0$ students degenerate into empty or near-empty outputs and do not constitute a valid transfer test (SI Table S26). Under the tested procedure, the readout pattern therefore does not transfer.

Competence extrapolation (exploratory).

Three fits (OLS linear, OLS quadratic, GP with RBF + white kernel) over the 42 unique non-released binaries (dev BLEU-4 0–9.8; 37 above 4.5, the five degenerate checkpoints at dev ≈ 0) all predict gaps at dev 13.38 far below the observed +10.24: OLS linear -1.84 (95% PI $[-2.67, -1.01]$), OLS quadratic -1.09 , GP -1.00 ($[-2.16, +0.17]$). Model sensitivity is substantial — the quadratic shifts to $+2.88$ on the 37 non-degenerate checkpoints and to $+0.23$ when distillation is excluded — while GP predictions stay negative with leave-one-family-out upper bounds below $+0.3$. We treat this as an exploratory visual guide (SI Sup. K), not evidence that competence alone cannot explain the reversal, since non-linear behavior inside the unobserved (9.8, 13.38) interval cannot be excluded.

Synthesis.

Across the five descriptive comparisons, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78). No reconstruction or distilled student reproduces either the +10.24 gap or this combination.

4.6 Checkpoint-targeted oracle stress tests (SI)

We treat evaluator-aware donor selection as a checkpoint-targeted oracle stress test, not a deployable SLP method: the unmasked diagonal advantage does not survive reference-overlap exclusion, and the white-box search budget (candidate-pool size) controls whether any decoded-BLEU margin is detectable. The full cross-checkpoint transfer matrix, candidate-pool sweep, and selector/evaluator-label permutation test are in SI Sup. H, which they share with the main estimands by design.

5 Discussion

The audit yields three findings organized around the question of whether the publicly documented recipe is sufficient to reproduce the released SLRTP2025 BT evaluator and its behavior on a constructed replay probe.

RQ1—Public-recipe insufficiency. The released evaluator achieves a combination of training-data fit and in-domain generalization that no checkpoint constructed from the public recipe reproduces. Fourteen recipe-conditioned reconstructions reach development-set free-decode BLEU 6.5–9.4 versus the released 13.38, and training-set teacher-forced token accuracy 55–60% versus 96.5% (Table 3). The training-pool free-decode readout is 78.8 BLEU (70.7% EM) versus a family max of 9.3. A small-sample overfit test (BLEU reaching 100.0 on 50–100 samples) and a unified-inference regression test (all vocabs byte-identical across checkpoints; released REC reproduced

at 12.78) exclude fundamental code-path defects as the explanation. Teacher-forced development-set NLL is actually *lower* for the reconstructions (2.63–2.68 vs. released 3.24), consistent with less overfitting, but their teacher-forced token accuracy is comparable (45.8–47.4% vs. 48.6%). The gap between released and reconstructed free-decode BLEU is therefore present in both training and development data and is already visible in teacher-forced token-level metrics, not only in free-decoding. These five converging descriptive comparisons (Table 7)—gap-competence, pass-through ratio, training-pool readout, frame-reversal loss, and the competence gate—all place the released evaluator far outside any constructible range. The insufficiency is not failure of the reconstruction code but failure of the public recipe to produce a checkpoint matching the released one’s fit to the training data.

RQ2—Non-reproduction of the stress-test ordering. The fourteen same-recipe reconstructions (shared architecture, tokenizer, manifest, selection rule; seeds 101–1405) give a seed-level PURE–REC gap prediction interval of $[-1.94, -0.46]$; the six pre-registered primary seeds alone yield $[-2.48, -0.28]$, all fourteen leave-one-seed-out folds keep the interval below zero, and a two-level seed $\times$ item bootstrap CI is $[-1.49, -0.93]$ (§4.2). Twenty-four heterogeneous-family checkpoints (config-faithful, step-faithful, confirmation, long-schedule, rescue, distillation students, and ladder fractions; all decoded with the canonical donor registry) have strictly negative gaps (range $[-2.01, -0.21]$ per family, $[-2.01, -0.77]$ for the twelve distillation students; SI Sup. D). All counts are run counts after SHA-256 deduplication: 43 decoded runs $\rightarrow$ 42 unique weight binaries, 38 non-degenerate runs (37 unique) all with strictly negative gaps (§3). Cross-evaluator re-decoding of the noisy-gloss retrieval pipeline output (a deployment-realistic analogue of TN-PURE-v1; 641 PHX-public keys, SI Table S32) yields a released-evaluator gap of -0.13 (noisy-gloss 12.65 vs. REC 12.78) and 38/38 negative gaps across non-degenerate evaluators. Because all constructible checkpoints are less competent than the released evaluator (decoded family max dev BLEU-4 9.8, training-log max 10.02, vs. 13.38), this non-reproduction is scoped to the public recipe and does not constitute a causal checkpoint-identity claim. Two local-robustness checks on the released checkpoint (Gaussian weight noise and fine-tuning from released weights, SI Sup. F) confirm the reversal is locally robust but probe only the vicinity of the released weights; OLS/GP extrapolation from the 42 unique non-released binaries is treated as exploratory visualization (SI Sup. K) because non-linear behavior inside the unobserved (9.8, 13.38) interval cannot be excluded. Our rebuilt USTC-MoE output (SI Table S33) is pipeline-faithful but not score-faithful (2.42 vs. official 12.06) and should not be taken as representative of the original competition entry; the noisy-gloss result provides the primary realistic-system evidence.

RQ3—Reference-frame sensitivity and donor-origin coupling. On the matched 461-item confidence=1 subset of Czehmann et al.’s Czehmann et al. (2025) human back-translations, reference replacement attenuates the gap by 90.5% ($+10.03 \rightarrow +0.95$, paired bootstrap CI $[7.44, 10.86]$ absolute, $[0.81, 1.01]$ ratio). Because the source-conditioned retriever selects donors by lexical overlap with the German source sentence, which is identical to the original reference text in PHX-public (§3.2), the attenuation reflects both reference-text information loss and disruption of the structural lexical coupling between the probe and the original reference. The asymmetric

PURE-vs-REC drop under reference replacement ($1.31\times$ BLEU-4; $1.29\times$ full-641) is numerically inconsistent with simple symmetric floor compression, but BLEU’s non-linearity and the coupling structure preclude a definitive floor-effect null-hypothesis test. Our contribution is applying the matched-subset protocol to the released evaluator and a frozen replay probe, not re-establishing reference defects that Czehmann et al. already documented. The donor-pool substitution (§4.3) reverses the gap sign (UNSEEN-PURE-v1 scores 8.51 vs. REC 12.78, gap -4.27), and the factorial interaction between pool origin and donor similarity ($+3.70$) confirms that seen donors benefit from both provenance and similarity; because seen donors remain more similar than unseen within strata, this is a provenance contrast, not an identified training-exposure effect.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-recipe reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal — teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479) — and whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025).

Value of the audit infrastructure despite the competence confound.

Because the competence-matched condition is unmet, RQ2’s non-reproduction result cannot be attributed to a causal checkpoint effect—it could reflect the competence gap, an unrecovered training mechanism, or their interaction. This constraints what the audit *infers* but not what it *provides*. The evidence infrastructure the audit contributes is independent of the competence confound and has lasting value for the language-resource and SLP communities. First, the provenance table (Table 2) and three-way data reconciliation establish a verified ground truth for any future PHX-public BT-evaluator study: the released pose materialization differs from the official PHOENIX-2014T split by 41 IDs, and the released evaluator’s manifest matches the former, not the latter. Second, the checkpoint registry with SHA-256 deduplication (SI Sup. D) demonstrates a standard for BT-evaluator auditing that prevents double-counting and enables exact third-party verification. Third, the unified-inference regression test (§3.4) provides a reusable protocol for isolating code-path defects from recipe insufficiency in any learned-metric reproducibility audit. Fourth, the leakage controls (pose permutation, label permutation, held-out signer; SI Sup. M) are an extensible diagnostic suite for learned evaluators on any dataset. Fifth, the single canonical materialization (SHA-256 9170a530) with its three-layer reproduction mechanism (single-command regenerability, per-script re-executability, and full retraining scriptability) is a concrete model for reproducible benchmarking infrastructure. These contributions do not depend on resolving the competence confound and are directly transferable to future audits of learned evaluation metrics on other language resources.

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: a released evaluator is a research artifact, not an infrastructure component. The released `config.yaml` alone was insufficient for 70 same-recipe runs to reach competence parity, and the readout pattern could not be localized to a training stage because intermediate checkpoints are unpublished. Benchmarks should release (i) the exact training command and seeds, (ii) data pipeline scripts beyond the manifest, (iii) a SHA-256 manifest of training tensors, (iv) intermediate checkpoints at least every 100 epochs, and (v) the fixed per-team pose outputs alongside the public scores; the current release provides only the Progressive Transformer baseline output, so rank-stability audits across independently trained evaluators cannot be reproduced by third parties on real competition systems.

Lesson 2: on template-dense benchmarks, retrieval entries benefit from shortcut-robustness disclosure. A donor-exclusion sweep is cheap and exposes the retrieval advantage on PHX-public’s template-dense register; leaderboard entries that retrieve from the training pool could attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results for transparency. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$; reconstructions: 0.93–1.12) and input-side pose permutation with output-equivariance verification (SI Sup. M) are two descriptive probes we found informative in this case study; whether they generalize to other benchmarks is untested.

Lesson 3: reference-coupling artifacts warrant a supplementary reference check. On PHX-public, the 90.5% attenuation under Czehmann et al. Czehmann et al. (2025) back-translations shows that rankings against a single speech-transcribed reference can diverge sharply from an independent sign-conditioned reference. Benchmarks on similarly template-dense corpora could adopt human back-translations as a supplementary reference set, follow the matched-subset protocol, and flag systems whose gap shifts between the two; the cost is one reference file and a paired bootstrap. Whether this generalizes beyond PHX-public’s lexical-coupling structure is untested.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2025); Alkain et al. (2026). The fixed nine-system stress-test set comprises replay, composition, weak generation, and random controls rather than competitive SLP systems; the public leaderboard (§4.4) offers only single-evaluator context; per-team pose outputs are not archived, and our two fixed-output cross-evaluator studies — the noisy-gloss retrieval pipeline and the official winner’s outputs rebuilt from its released intermediates — decode at or below REC under all 38 non-degenerate evaluators, but the USTC-MoE rebuild is not score-faithful to the archived submission (2.42 vs. the official leaderboard score of 12.06), so no causal claim about evaluator retrieval-favoring follows. All 67 non-holdout checkpoints are less competent than the released evaluator (decoded family max dev BLEU-4 9.8, training-log max 10.02, vs. 13.38); the original training history is unpublished, so competence-matched replication from the public recipe was not possible, and local-robustness checks (Gaussian weight noise and fine-tuning

from the released checkpoint, SI Sup. F) confirm the reversal is not a fragile singular point but do not narrow the unobserved competence interval. Saunders et al. Saunders et al. (2021) limited their rank-stability claim to similar configurations, which our competence-unmatched family does not directly test. The mechanism is not identifiable from public artifacts: neither recipe-conditioned reconstruction, rescue sweeps, config-faithful retraining, nor any tested distillation procedure transfers the property; the donor-pool substitution decomposition is path-dependent, and pool-origin contrasts are descriptive associations, not identified training-exposure effects. The human back-translations are a single-annotator derivative resource without second-annotator adjudication. Czehmann et al.’s confidence flags partition the full 641 items into 461 confidence=1, 121 confidence=0.5, and 59 confidence=0.0 tiers. Restricting to confidence=1 (our matched-subset protocol) is conservative for the BLEU analysis—lower-confidence items have shorter, more fragmented back-translations that would inflate the floor effect—but it discards 180 items (28%), and a single annotator’s confidence ratings are not independently validated. We cannot estimate inter-annotator agreement or quantify how a second independent back-translator would shift the attenuation. Qualitatively, the direction of the attenuation is stable across confidence tiers (confidence=0.5: gap $+13.3 \rightarrow +4.1$, 69% attenuation; confidence=0.0: gap $+6.7 \rightarrow +0.2$, 97% attenuation; SI Table S31), but the quantitative attenuation fraction varies with the subset, and the small sample sizes at lower confidence tiers preclude reliable estimation. The human-reference protocol should be viewed as a single-annotator case study whose quantitative attenuation is conditional on one signer’s interpretation; its qualitative conclusion—that reference replacement substantially reduces the gap—is robust to subset choice, but the precise 90.5% estimate should not be taken as a population parameter. Finally, the reversal is a decoded-BT-text-metric reversal, not a sign-language quality reversal: without target-conditioned DGS signer judgment or a validated sign-native pose metric, we cannot determine whether the retrieved donor motion conveys target-equivalent meaning, and we explicitly do not claim that donor motion fails to express the target, nor that the reversal indicates dataset contamination or pose memorization.

6 Conclusion

This public-recipe sufficiency audit reports three findings, organized by research question, on a +10.24 BLEU-point retrieval-recorded-poses reversal under the released SLRTP2025 BT evaluator on PHX-public.

For RQ1, the publicly documented training recipe does not reproduce the released evaluator’s competence. Fourteen recipe-conditioned reconstructions reach development-set free-decode BLEU 6.5–9.4 (vs. released 13.38) and training-set teacher-forced token accuracy 55–60% (vs. released 96.5%); a small-sample overfit test (BLEU 100.0 on 50–100 samples) and a unified-inference regression test exclude fundamental code-path defects. The released evaluator’s training-pool readout (78.8 BLEU, 70.7% EM) is $8.5\times$ above the family maximum.

For RQ2, the stress-test ordering does not reproduce: 38 non-degenerate decoded runs (37 unique weight binaries after SHA-256 deduplication) all yield negative

PURE-REC gaps (reconstruction range $[-1.85, -0.80]$; heterogeneous-family ranges per family in SI Sup. D). Cross-evaluator re-decoding of a realistic noisy-gloss retrieval output yields 38/38 negative gaps across non-degenerate evaluators.

For RQ3, reference replacement attenuates the gap by 90.5% on the matched 461-item confidence=1 subset (ratio CI $[0.81, 1.01]$), and donor-origin substitution reverses its sign. The source-conditioned retriever shares surface-level vocabulary with the original reference (§3.2), so attenuation reflects both reference-text properties and disruption of this lexical coupling.

The scope is bounded. Local-robustness checks confirm the reversal is not a fragile singular point near the released checkpoint, but they probe only its vicinity; OLS/GP extrapolation from the 42 unique non-released binaries is exploratory because non-linear behavior inside the unobserved competence interval (9.8, 13.38) cannot be excluded. Because competence-matched replication was not possible from public artifacts, the findings evidence public-recipe insufficiency for reproducing the released evaluator and its behavior on a constructed replay probe—not a causal checkpoint effect, an identified memorization mechanism, or general leaderboard-rank instability.

Declarations

Funding.

[FUNDING-PLACEHOLDER: list grants or state that no funds, grants, or other support were received.]

Conflict of interest.

The authors declare no competing interests.

Ethics approval.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and published human back-translations Czehmann et al. (2025); no new human-subjects data was collected.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the artifact at <https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md>; a persistent DOI-bearing release will be deposited upon acceptance. The supplementary material ([supplementary.pdf](#)) contains all SI appendices. Upstream datasets are used under their published licences (PHOENIX-2014T and the human back-translations under CC BY-NC-SA 4.0; CSL-Daily and SLRTP2025 under their academic licences); pose tensors may retain signer identity, so we release no per-signer raw poses.

Author contribution.

[AUTHOR-CONTRIBUTION-PLACEHOLDER]

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact. Each appendix is annotated with the question it answers.

- Sup. A: Missing Test ID Sensitivity — does the absent test item change the headline gap? (`sec:appendix_missing_id`)
- Sup. B: Per-seed Extension Cells 707–1405 — does the prediction interval hold beyond the six primary seeds?
- Sup. C: Analysis Provenance — evidentiary role and multiplicity handling of each analysis. (`sec:provenance`)
- Sup. D: Canonical Checkpoint Registry — what was trained, and which families enter which claim? (`tab:checkpoint_registry`)
- Sup. E: Donor-Pool Substitution Decomposition — how much of the seen-pool advantage is pool-origin vs. similarity? (`sec:appendix_decomposition`, `tab:path_sensitivity`)
- Sup. F: Config-Faithful and Perturbation Details — protocol-mismatch check, weight-noise sweep, epoch-checkpoint decoupling search. (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`)
- Sup. G: Cluster-Aware Bootstrap — does the gap survive resampling by signer, broadcast-show, and date? (`sec:appendix_cluster_bootstrap`)
- Sup. H: Supplementary Oracle Diagnostics — is the evaluator-aware advantage robust to reference-overlap exclusion? (`sec:appendix_oracle`)
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion — does the reversal persist on a clean template-holdout split and under donor-text exclusion? (`sec:appendix_template_boundary`)
- Sup. J: Transcript-Slot Diagnostic Details — do slot-level scores reproduce the BLEU reversal? (`sec:appendix_slot`)
- Sup. K: Competence Extrapolation and Per-Item Gap Decomposition — could competence in the unobserved interval explain the reversal, and which item features drive the gaps? (`sec:appendix_extrapolation`)
- Sup. L: CSL-Daily Boundary Control — does the readout pattern generalize beyond PHX-public? (`sec:csl-daily`)
- Sup. M: Leakage Sanity Checks — can the train-pool readout be explained by implementation shortcuts? (`sec:appendix_leakage`)
- Sup. N: Pre-Exclusion Materialization Sensitivity — how sensitive are the headline numbers to the donor-exclusion rule? (`sec:appendix_materialization`)
- Sup. O: BLEURT-Substitute Validation — does a learned metric corroborate the BLEU finding under reference replacement? (`sec:appendix_bleurt`)
- Sup. P: Main-Text Tables (SI Table S22–S33) — decoded-metric sensitivity, floor effect, cross-metric gaps, leaderboard, distillation ladder, checkpoint registry, protocol summary, rescue sweep, exposure estimands, reference sensitivity, and cross-evaluator re-decoding of noisy-gloss and USTC-MoE outputs. (`sec:appendix_moved_tables`)
- Sup. Q: Additional Protocol Details — tie-breaking, scorer equivalence, metric implementations, audit-design, reference-frame, and donor-origin figures (SI Fig. S4–S6). (`sec:appendix_moved_protocol`)
- Sup. R: Reconstruction Sanity Checks (SI Table S35) — small-sample overfit test, unified inference regression, and per-checkpoint teacher-forced vs. free-decode diagnostics ruling out a systematic code-path defect. (`sec:appendix_sanity`)

Full checkpoint registry in SI Sup. D.

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